ADVANCED SPLUNK ENTERPRISE SECURITY

Beginner-Friendly Security Operations & SIEM Course

90 Hours (4.5 months) • Hands-on Labs • Mission Control • Detection Engineering • Risk-Based Alerting • Threat Investigation

TOTAL COURSE DURATION: 90 HOURS

Practical / Hands-On

Recommended delivery: 30 sessions × 3 hours (or equivalent instructor-led schedule). Module hours include instructor explanation, demonstrations, guided ES labs, investigation exercises, troubleshooting and short assessments.

Course Overview

This course introduces Splunk Enterprise Security (ES) from a beginner's perspective and gradually moves learners toward practical SOC investigation and detection skills. Learners first understand Splunk and SIEM fundamentals, then work with Enterprise Security workflows such as Mission Control, detections, findings, investigations, security content, data models, risk analysis and threat intelligence.

The course is designed for a lab environment using an available Splunk Enterprise Security trial/evaluation installation. Exact ES features can vary by Splunk ES version, license, installed content updates and lab data. The instructor should verify the feature set shown by the trial before the corresponding lab.

What Participants Will Learn

- Understand how Splunk Enterprise Security supports SOC operations.
- Navigate the Enterprise Security application and Mission Control.
- Understand findings, finding groups, investigations and analyst workflows.
- Use SPL to investigate security events.
- Understand CIM and data-model-driven security analytics.
- Create and tune detections for common SOC use cases.
- Understand risk-based alerting (RBA) and risk scoring.
- Use threat intelligence and observable context in investigations.
- Map detections to MITRE ATT&CK concepts.
- Build dashboards and analyst views.
- Troubleshoot data, detection and ES-content issues.
- Complete an end-to-end SOC investigation.

Prerequisites

- Basic computer and networking knowledge
- Basic understanding of cybersecurity concepts
- Basic SPL knowledge is helpful but not mandatory
- No previous Splunk Enterprise Security experience required
- A Splunk Enterprise + Enterprise Security trial/evaluation lab
- Sample Windows/Linux/security logs for exercises

Recommended Lab Environment

- Splunk Enterprise with a valid Enterprise Security trial/evaluation entitlement
- Splunk Enterprise Security application compatible with the platform version
- Windows and/or Linux test VM for generating logs
- Optional Universal Forwarder
- Optional Sysmon for richer Windows telemetry
- Sample authentication, endpoint, DNS, web and network logs
- MITRE ATT&CK reference material
- Instructor-created security scenarios

Important Current ES Concepts

Current Splunk Enterprise Security documentation centers the analyst workflow around Mission Control, where findings and investigations can be triaged, investigated and responded to. ES also supports detections, threat-intelligence context, response plans and risk-based alerting.

Risk-based alerting aggregates risk signals associated with entities such as users and systems so that multiple lower-level observations can contribute to higher-context findings. Splunk documentation describes this as a way to correlate behavior across time, data sources and MITRE ATT&CK context.

Suggested Course Roadmap

Phase	Modules	Learning Focus
Foundation	01–04	Splunk/SIEM, ES architecture, navigation and data
SOC Investigation	05–08	Mission Control, findings, investigations and SPL
Detection Engineering	09–12	CIM, data models, detections and tuning
Risk & Intelligence	13–15	RBA, threat intelligence and ATT&CK
Operations & Capstone	16–18	Dashboards, troubleshooting and end-to-end SOC project

Module Duration Summary

Module 01 — Splunk & SIEM Foundations (4 Hours)

Module Focus: Build the minimum Splunk knowledge needed before entering Enterprise Security.

Theory Concepts

- What is SIEM?
- Splunk as a security analytics platform
- Events, fields, timestamps and metadata
- Indexes, sourcetypes and hosts
- SPL search fundamentals
- Security telemetry sources
- SIEM workflow: collect → normalize → detect → investigate → respond
- SOC analyst roles and responsibilities

Hands-on Lab

Lab Objective: Explore a Splunk environment and perform basic security searches.

Lab Activities

1. Open Search & Reporting and inspect the lab data.
2. Identify indexes, sourcetypes and hosts.
3. Search for authentication events.
4. Filter events by user, source IP and time.
5. Create a simple failed-login search.
6. Save the search for later use.

Splunk / Lab Tools

- Splunk Enterprise
- Search & Reporting
- Sample security logs

Expected Outcome

Participant understands the Splunk foundation required for ES.

Module 02 — Enterprise Security Overview & Architecture (4 Hours)

Module Focus: Understand what ES adds to the Splunk platform and how its major security components fit together.

Theory Concepts

- Purpose of Splunk Enterprise Security
- Security domains: access, endpoint, identity, network and threat
- ES content and configuration
- Mission Control
- Detections and findings
- Investigations
- Security content and analytics
- CIM and data models
- Risk framework overview

Hands-on Lab

Lab Objective: Map the major ES components to a simple SOC architecture.

Lab Activities

7. Open the Enterprise Security application.
8. Identify Mission Control, Analytics, Security Content and Configure areas available in the lab.

9. Trace a security event from raw data toward a detection/finding.
10. Create a simple architecture diagram.
11. Explain which components are analyst-facing versus administration/content-facing.

Splunk / Lab Tools

- Splunk Enterprise Security
- Splunk Web

Expected Outcome

Participant can explain ES architecture in simple SOC terms.

Module 03 — Enterprise Security Data & CIM Fundamentals (5 Hours)

Module Focus: Learn why normalized security data is important for reusable detections and analytics.

Theory Concepts

- Common Information Model (CIM) concept
- Data models and datasets
- Normalized fields
- Tags and event types
- Authentication, Endpoint, Network and Web data concepts
- Data model acceleration concept
- Why inconsistent fields break detections
- Data onboarding validation

Hands-on Lab

Lab Objective: Validate whether sample security data is suitable for ES analytics.

Lab Activities

12. Identify a raw authentication dataset.
13. Compare its fields with expected normalized security fields.
14. Review relevant CIM/data-model concepts in the ES environment.
15. Run a simple data-model search where available.
16. Identify missing or inconsistent fields.
17. Document normalization improvements.

Splunk / Lab Tools

- Splunk Enterprise Security
- CIM Add-on
- Data Models
- SPL

Expected Outcome

Participant understands why CIM/data models are important to ES detections.

Module 04 — Enterprise Security Navigation & Analyst Workflow (4 Hours)

Module Focus: Become comfortable with the ES interface before investigating real scenarios.

Theory Concepts

- Mission Control
- Analyst queue
- Findings and finding groups
- Investigation records
- Security domains
- Severity and status
- Timeline and event context
- Disposition concepts

Hands-on Lab

Lab Objective: Perform a complete beginner investigation workflow using instructor-generated findings.

Lab Activities

18. Open Mission Control.
19. Filter and sort available findings.
20. Open a finding and review its details.
21. Inspect related entities, events and context.
22. Start or update an investigation.
23. Record evidence and analyst notes.
24. Set an appropriate training disposition and close the investigation.

Splunk / Lab Tools

- Mission Control
- Enterprise Security sample content

Expected Outcome

Participant can navigate ES and follow a structured analyst workflow.

Module 05 — Security Events & SPL Investigation (5 Hours)

Module Focus: Use SPL to investigate common security scenarios behind ES findings.

Theory Concepts

- Time-range investigation
- Authentication analysis
- Source/destination analysis
- stats, timechart, eval and where
- Field extraction
- Event correlation
- Evidence preservation
- Pivoting from summary to raw events

Hands-on Lab

Lab Objective: Investigate a simulated suspicious-login scenario.

Lab Activities

25. Find repeated authentication failures.
26. Identify the affected user and source IP.
27. Look for a subsequent successful authentication.
28. Build a timeline.
29. Search for related activity from the same user/IP.
30. Summarize evidence and determine what additional data is needed.

Splunk / Lab Tools

- Splunk SPL
- Windows/Linux authentication logs

Expected Outcome

Participant can use SPL to support an ES investigation.

Module 06 — Mission Control: Findings & Investigations (5 Hours)

Module Focus: Learn how analysts triage, investigate and document security findings.

Theory Concepts

- Finding lifecycle
- Finding groups
- Investigations
- Analyst queue
- Entities and threat objects
- Notes and evidence
- Investigation status
- Disposition and closure
- Collaboration concepts

Hands-on Lab

Lab Objective: Work through a simulated SOC case from finding to closure.

Lab Activities

31. Select a training finding.
32. Review the detection and evidence.
33. Group related findings when appropriate.
34. Create/update an investigation.
35. Add notes and evidence.
36. Track investigative status.
37. Choose a training disposition and document the reasoning.

Splunk / Lab Tools

- Mission Control
- Training findings

Expected Outcome

Participant can manage a basic investigation in Mission Control.

Module 07 — Security Content & Detection Concepts (5 Hours)

Module Focus: Understand how ES security content turns analytics into actionable findings.

Theory Concepts

- Detection concept
- Scheduled/correlation searches
- Event-based versus finding-based detection
- Detection metadata
- Severity and security domain
- Annotations
- Threat objects
- Detection testing and validation

Hands-on Lab

Lab Objective: Inspect existing detections and build a simple training detection.

Lab Activities

38. Open the available security-content/detection area.
39. Inspect an existing detection and its SPL.
40. Identify schedule, time range and output behavior.
41. Create a simple authentication anomaly detection for lab data.
42. Test it with controlled events.
43. Document detection logic and expected result.

Splunk / Lab Tools

- Security Content
- Detection Editor
- SPL

Expected Outcome

Participant can explain how detections produce actionable security findings.

Module 08 — Common SOC Use Cases in ES (5 Hours)

Module Focus: Translate common security scenarios into searches and detections.

Theory Concepts

- Brute-force authentication
- Suspicious successful login
- Privilege escalation indicators
- Malware/endpoint alerts
- Suspicious outbound traffic
- Internal scanning
- Unusual account activity
- Data access anomalies
- Use-case requirements and evidence

Hands-on Lab

Lab Objective: Create a small set of beginner SOC detections.

Lab Activities

44. Create a failed-login detection.
45. Create a repeated-source-IP detection.
46. Create a suspicious process/event search using available logs.
47. Create a simple internal scanning search where suitable data exists.
48. Map each use case to relevant data sources and expected false positives.
49. Document tuning ideas.

Splunk / Lab Tools

- Splunk ES
- SPL
- Windows/Linux/network sample data

Expected Outcome

Participant can turn common SOC questions into practical ES detection logic.

Module 09 — Detection Engineering & Tuning (5 Hours)

Module Focus: Learn how to improve detections so analysts receive useful, explainable results.

Theory Concepts

- Detection quality
- False positives
- Thresholds and time windows
- Allow lists/exclusions
- Macros and reusable filters
- Testing and validation
- Detection coverage
- Change management
- Performance considerations

Hands-on Lab

Lab Objective: Take a noisy training detection and tune it.

Lab Activities

50. Run the initial detection and record results.
51. Identify benign activity causing noise.
52. Create a controlled exclusion/allow-list approach.
53. Adjust threshold or time range.
54. Re-test against both benign and suspicious training events.
55. Record before/after behavior and tuning rationale.

Splunk / Lab Tools

- Detection Editor
- SPL
- Lookups
- Macros

Expected Outcome

Participant can tune a detection without simply disabling it.

Module 10 — Data Models & Security Analytics (5 Hours)

Module Focus: Use ES data models to understand normalized analytics and security dashboards.

Theory Concepts

- Data-model structure
- Datasets and constraints
- CIM mapping
- Data-model acceleration
- Pivot concepts
- Datamodel command
- tstats concept
- Performance benefits
- Troubleshooting data-model coverage

Hands-on Lab

Lab Objective: Compare raw-event searches with data-model-driven searches.

Lab Activities

56. Identify a relevant ES/CIM data model.
57. Review its datasets and expected fields.
58. Run a basic data-model query where supported.
59. Compare results with raw-event SPL.
60. Identify missing data-model coverage.
61. Document what needs to be fixed upstream.

Splunk / Lab Tools

- CIM
- Data Models
- tstats
- datamodel

Expected Outcome

Participant understands how normalized data powers scalable ES analytics.

Module 11 — Security Dashboards & Analytics (4 Hours)

Module Focus: Use ES analytics views to understand security posture and investigation trends.

Theory Concepts

- Security analytics concepts
- Dashboards versus investigations
- Time-series analysis
- Top entities
- Authentication trends
- Endpoint/network visibility
- Dashboard filtering
- Drilldowns and analyst usability

Hands-on Lab

Lab Objective: Create a beginner security-monitoring dashboard using available ES/Splunk data.

Lab Activities

62. Create panels for authentication failures.
63. Add top source IPs/users.
64. Add an event trend over time.
65. Add a suspicious-activity panel.
66. Add a time-range filter.
67. Configure a drilldown to detailed events.
68. Document what each panel tells an analyst.

Splunk / Lab Tools

- Splunk dashboards
- Dashboard Studio/Simple XML
- SPL

Expected Outcome

Participant can build a practical security dashboard for SOC monitoring.

Module 12 — Risk-Based Alerting (RBA) Fundamentals (5 Hours)

Module Focus: Understand how ES can combine multiple risk signals into higher-context findings.

Theory Concepts

- Risk events and risk scores
- Risk objects and entities
- Risk index concept
- Risk factors
- Risk incident rules
- Risk findings
- Thresholds and time windows
- Behavior over time
- Why RBA can reduce alert noise

Hands-on Lab

Lab Objective: Build a controlled RBA scenario using synthetic security events.

Lab Activities

69. Identify a training user or host as the risk entity.
70. Create or inspect a low-level risk signal.
71. Assign a training risk score.
72. Generate a second related risk signal.
73. Review how aggregated risk can contribute to a finding.
74. Inspect the risk timeline/analysis view available in the lab.
75. Document how the combined signals tell a stronger story.

Splunk / Lab Tools

- Risk Analysis
- Risk index
- Risk-based detections
- SPL

Expected Outcome

Participant understands the core RBA workflow and why multiple weak signals can form a stronger investigation.

Module 13 — Risk Analysis & Threat Prioritization (4 Hours)

Module Focus: Use risk context to prioritize entities and investigate connected behavior.

Theory Concepts

- Risk aggregation
- Risk modifiers
- Entity-centric investigation
- Risk timeline
- High-risk users/systems
- Cross-source correlation
- Risk threshold tuning
- Context and investigation priority

Hands-on Lab

Lab Objective: Investigate a simulated compromised-user scenario using risk context.

Lab Activities

76. Review risk events associated with the training user.
77. Identify contributing detections.
78. Review the timeline across multiple event types.
79. Determine which evidence increases confidence.
80. Identify benign contributors.
81. Write a concise risk-based investigation summary.

Splunk / Lab Tools

- Risk Analysis
- Mission Control
- SPL

Expected Outcome

Participant can investigate a security story across multiple signals rather than a single alert.

Module 14 — Threat Intelligence & Observable Investigation (4 Hours)

Module Focus: Learn how threat intelligence can enrich security investigations.

Theory Concepts

- Threat indicators and observables
- IP, domain, URL and hash concepts
- Threat lists
- Threat matching
- Indicator context
- False-positive considerations
- Threat-intelligence freshness

- Enrichment during investigation

Hands-on Lab

Lab Objective: Enrich a controlled set of synthetic observables.

Lab Activities

82. Create a small instructor-provided threat-indicator dataset.
83. Load/use the dataset in the lab's supported threat-intelligence workflow.
84. Search for matches in training logs.
85. Review matched observable context.
86. Validate matches before treating them as malicious.
87. Document enrichment limitations and false-positive handling.

Splunk / Lab Tools

- Enterprise Security threat-intelligence features available in trial
- CSV training indicators
- SPL

Expected Outcome

Participant can use threat intelligence as supporting evidence rather than treating a match as automatic proof.

Module 15 — MITRE ATT&CK & Detection Coverage (4 Hours)

Module Focus: Connect ES detections to adversary behaviors and identify coverage gaps.

Theory Concepts

- MITRE ATT&CK tactics and techniques
- Detection mapping
- Technique metadata
- Threat stories
- Coverage visualization concepts
- Detection gaps
- Use-case prioritization
- Evidence versus assumption

Hands-on Lab

Lab Objective: Map a small detection library to MITRE ATT&CK.

Lab Activities

88. Select 5–8 training detections.
89. Identify the relevant ATT&CK technique for each.
90. Record the data source and detection logic.
91. Create a simple coverage matrix.
92. Identify techniques with no current detection.
93. Propose one additional lab detection for a coverage gap.

Splunk / Lab Tools

- MITRE ATT&CK
- Splunk ES detections
- Coverage matrix

Expected Outcome

Participant can explain how ATT&CK mapping helps organize detection coverage.

Module 16 — Incident Response & Response Plans (4 Hours)

Module Focus: Move from detection to structured response using investigation procedures.

Theory Concepts

- Incident-response lifecycle
- Triage
- Containment concepts
- Evidence collection
- Response plans
- Investigation types
- Analyst handoff
- Disposition and closure
- Lessons learned

Hands-on Lab

Lab Objective: Run a complete simulated incident-response case.

Lab Activities

94. Receive a simulated finding group.
95. Create an investigation.
96. Follow a response checklist/plan.
97. Collect supporting endpoint and network evidence.
98. Record actions and findings.
99. Document containment recommendations.
100. Close the case with lessons learned.

Splunk / Lab Tools

- Mission Control
- Response plans where available
- Wazuh/Windows/Linux logs as supporting data

Expected Outcome

Participant can connect ES investigation with a structured incident-response process.

Module 17 — ES Administration, Content Health & Troubleshooting (4 Hours)

Module Focus: Introduce the administrative tasks needed to keep ES useful and reliable.

Theory Concepts

- ES configuration areas
- Security content updates
- Data-model health
- Detection scheduling
- Search performance

- License/data-volume awareness
- Content dependencies
- Permissions
- Troubleshooting workflow

Hands-on Lab

Lab Objective: Diagnose a simulated ES problem.

Lab Activities

101. Inspect ES configuration and content status.
102. Identify a detection that is not producing expected results.
103. Check data availability and fields.
104. Review relevant internal logs/health information.
105. Identify whether the issue is data, SPL, schedule, permissions or content.
106. Document root cause and remediation.

Splunk / Lab Tools

- Splunk Web
- Monitoring Console
- _internal logs
- ES configuration

Expected Outcome

Participant can troubleshoot common beginner-level ES problems systematically.

Module 18 — Advanced Investigation & Threat Hunting (5 Hours)

Module Focus: Use ES and SPL to proactively search for suspicious behavior not yet represented by an obvious finding.

Theory Concepts

- Threat-hunting hypothesis
- Baseline behavior
- Rare events
- Anomaly indicators
- User and asset pivots
- Cross-source correlation
- Hunt documentation
- Hypothesis validation

Hands-on Lab

Lab Objective: Perform a guided threat hunt against synthetic security telemetry.

Lab Activities

107. Start with a hypothesis such as unusual authentication behavior.
108. Identify relevant data sources.
109. Search for rare or unusual patterns.
110. Pivot from user to host/IP and time.
111. Compare with baseline activity.
112. Document supporting and contradicting evidence.
113. Create a candidate detection from the hunt.

Splunk / Lab Tools

- Splunk ES
- SPL
- Risk Analysis
- Mission Control

Expected Outcome

Participant can perform a repeatable threat-hunting workflow and turn findings into detection ideas.

Module 19 — Final Capstone — Mini SOC with Splunk Enterprise Security (9 Hours)

Module Focus: Combine the course skills into an end-to-end SOC project.

Theory Concepts

- Project scoping
- Data onboarding
- CIM/data validation
- Detection engineering
- Mission Control workflow
- RBA
- Threat intelligence
- ATT&CK mapping
- Dashboarding
- Incident response
- Reporting

Hands-on Lab

Lab Objective: Build and demonstrate a small ES-based SOC monitoring and investigation environment.

Lab Activities

114. Define a fictional organization's security requirements.
115. Onboard at least two or three training datasets.
116. Validate fields/CIM coverage where applicable.
117. Create 5–8 practical detections.
118. Map selected detections to ATT&CK.
119. Create an RBA scenario using synthetic risk events where supported.
120. Build a security dashboard.
121. Generate a controlled incident.
122. Investigate it through Mission Control.
123. Produce a technical investigation report and executive summary.

Splunk / Lab Tools

- Splunk Enterprise
- Splunk Enterprise Security trial/evaluation
- Windows/Linux security logs
- Optional Universal Forwarder
- Synthetic threat-intelligence data

Expected Outcome

Participant completes a portfolio-ready ES SOC project demonstrating detection, investigation and risk-analysis skills.

Recommended Practical Use Cases

- Multiple failed logins followed by successful authentication
- Suspicious outbound connection from an endpoint
- Internal network/service scanning
- Privilege-related activity
- Suspicious process execution
- Known training IOC match
- Compromised-user behavioral sequence
- Repeated low-severity signals aggregated through risk
- Detection tuning and false-positive reduction
- End-to-end incident investigation

Final Assessment

- 30% — SPL and investigation exercises
- 20% — Detection engineering and tuning
- 15% — Mission Control investigation
- 15% — RBA / risk-analysis exercise
- 20% — Final Mini-SOC capstone and presentation

Lab Safety & Training Rules

All practical activities should use instructor-provided, synthetic, or explicitly authorized data and systems. Students should not test third-party systems, real credentials, production infrastructure or external targets. Threat-intelligence indicators used in exercises should be synthetic or instructor-approved whenever possible.

Official Documentation Alignment

This curriculum is designed around current Splunk Enterprise Security concepts documented by Splunk, including Mission Control, findings and investigations, detections, data models/CIM, threat intelligence and risk-based alerting.

- Mission Control supports analyst triage, investigations, timelines and response workflows.
- Enterprise Security detections can generate findings from security data and third-party alerts.
- ES uses CIM/data models for normalized security analytics and detection content.
- Risk-based alerting aggregates risk signals associated with entities over time to provide higher-context findings.
- Finding-based detections can use risk and notable data to create finding groups for investigation.

Course Positioning

SPLUNK ENTERPRISE SECURITY

From SIEM Fundamentals → SOC Investigation → Detection Engineering → Risk-Based Alerting

Splunk Enterprise Security — Beginner-Friendly Hands-on Training

Module	Module Title	Hours
01	Splunk & SIEM Foundations	4
02	Enterprise Security Overview & Architecture	4
03	Enterprise Security Data & CIM Fundamentals	5
04	Enterprise Security Navigation & Analyst Workflow	4
05	Security Events & SPL Investigation	5
06	Mission Control: Findings & Investigations	5
07	Security Content & Detection Concepts	5
08	Common SOC Use Cases in ES	5
09	Detection Engineering & Tuning	5
10	Data Models & Security Analytics	5
11	Security Dashboards & Analytics	4
12	Risk-Based Alerting (RBA) Fundamentals	5
13	Risk Analysis & Threat Prioritization	4
14	Threat Intelligence & Observable Investigation	4
15	MITRE ATT&CK & Detection Coverage	4
16	Incident Response & Response Plans	4
17	ES Administration, Content Health & Troubleshooting	4
18	Advanced Investigation & Threat Hunting	5
19	Final Capstone — Mini SOC with Splunk Enterprise Security	9
	TOTAL COURSE DURATION	90

Planning note: The 90 hours reflect the existing 19-module curriculum and its hands-on investigation, detection-engineering, RBA, threat-intelligence, threat-hunting and capstone workload. The ES trial/evaluation availability is a lab-environment constraint, not the teaching-hour count. Feature availability can vary by ES version, license and installed content.